

Employee Management System Using AWS Secrets Manager, AWS KMS and AWS S3

Project Overview

The Employee Management System is a cloud-enabled web application developed using Node.js, Express.js, MongoDB, and AWS services. The purpose of this application is to demonstrate how modern organizations can securely manage employee information while following cloud security best practices.

The application allows administrators to authenticate into the system, add employee records, upload employee profile photographs, store sensitive employee information, and automatically send welcome emails to newly onboarded employees. The project integrates multiple AWS services including AWS Secrets Manager, AWS Key Management Service (KMS), Amazon S3, and IAM to provide a secure and scalable solution.

Business Scenario

In many organizations, Human Resources departments manage large amounts of employee information. This information includes personal details, employee photographs, salary information, email addresses, and departmental data.

Several security challenges arise when handling such information.

Employee salary information is highly sensitive and should not be stored in plain text. If the database is compromised, confidential salary information could be exposed.

Application credentials such as database passwords, JWT secrets, SMTP passwords, and administrative login credentials should never be hardcoded inside source code because they can be exposed through code repositories or unauthorized access.

Employee profile photographs require scalable storage that can handle large amounts of data without increasing application server storage requirements.

To address these challenges, the Employee Management System leverages AWS cloud services to securely store credentials, encrypt sensitive information, and manage employee assets.

Solution Architecture

The application is deployed on an Amazon EC2 instance running a Node.js web application. The application communicates with AWS Secrets Manager to retrieve sensitive credentials, AWS KMS to encrypt and decrypt salary information, Amazon S3 to store employee profile photographs, and MongoDB to store employee records.

When an employee is added, the salary is encrypted using AWS KMS before being stored in MongoDB. Employee photographs are uploaded to Amazon S3 and only the object URL is stored inside the database.

When viewing employee details, the application retrieves the encrypted salary from MongoDB and uses AWS KMS to decrypt it before displaying it to authorized users.

AWS Services Used

AWS Secrets Manager

AWS Secrets Manager is used to securely store and manage application secrets. The application retrieves all secrets during startup and stores them in memory for efficient access.

The following secrets are managed using AWS Secrets Manager:

- MongoDB credentials
- Application admin credentials
- JWT signing secret
- SMTP email credentials

This approach eliminates hardcoded credentials and follows security best practices.

AWS Key Management Service (KMS)

AWS KMS is used to encrypt and decrypt employee salary information.

Whenever an employee is created, the salary value entered by the administrator is sent to AWS KMS for encryption. The encrypted ciphertext returned by KMS is stored in MongoDB instead of the original salary value.

When employee details are viewed, the encrypted value is retrieved from MongoDB and sent to AWS KMS for decryption.

This ensures that sensitive salary information remains protected while stored in the database.

Amazon S3

Amazon S3 is used to store employee profile photographs.

Instead of storing image files on the application server, employee photos are uploaded directly to an S3 bucket. The application stores only the image URL and object key inside MongoDB.

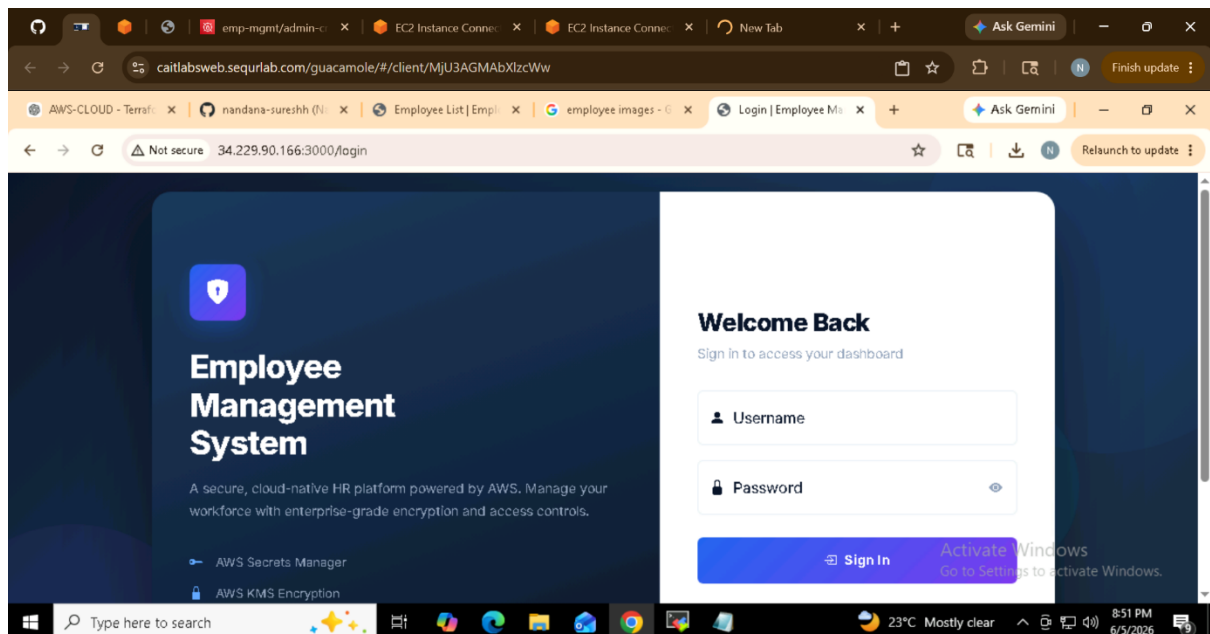
This approach improves scalability and reduces storage requirements on the application server.

IAM Roles

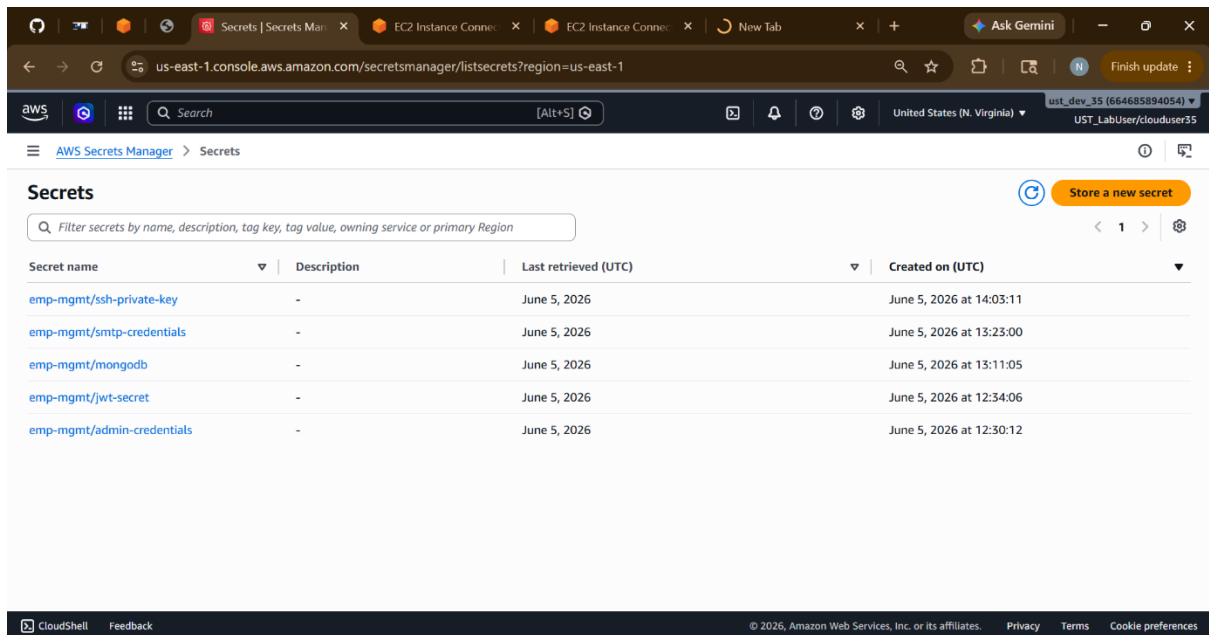
IAM Roles are used to provide secure access between the EC2 instance and AWS services.

The application EC2 instance uses an IAM Role with permissions to access Secrets Manager, KMS, and S3. This removes the need to store AWS access keys inside the application.

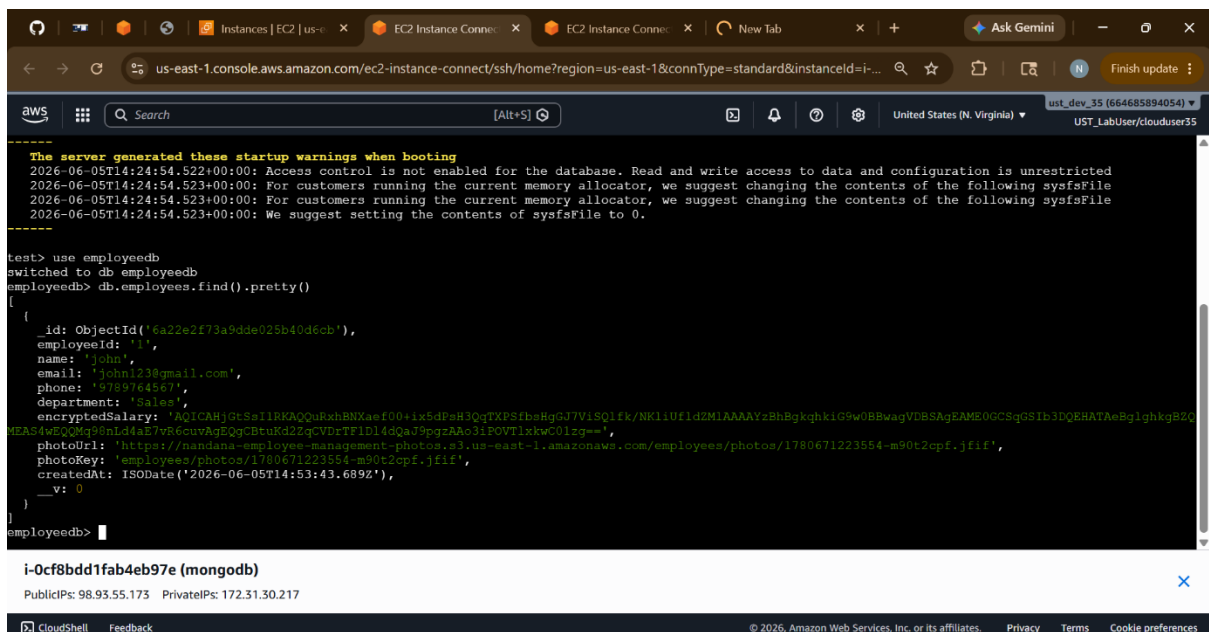
Developed a monolithic Employee Management System using Node.js, Express.js and MongoDB integrated with AWS services.



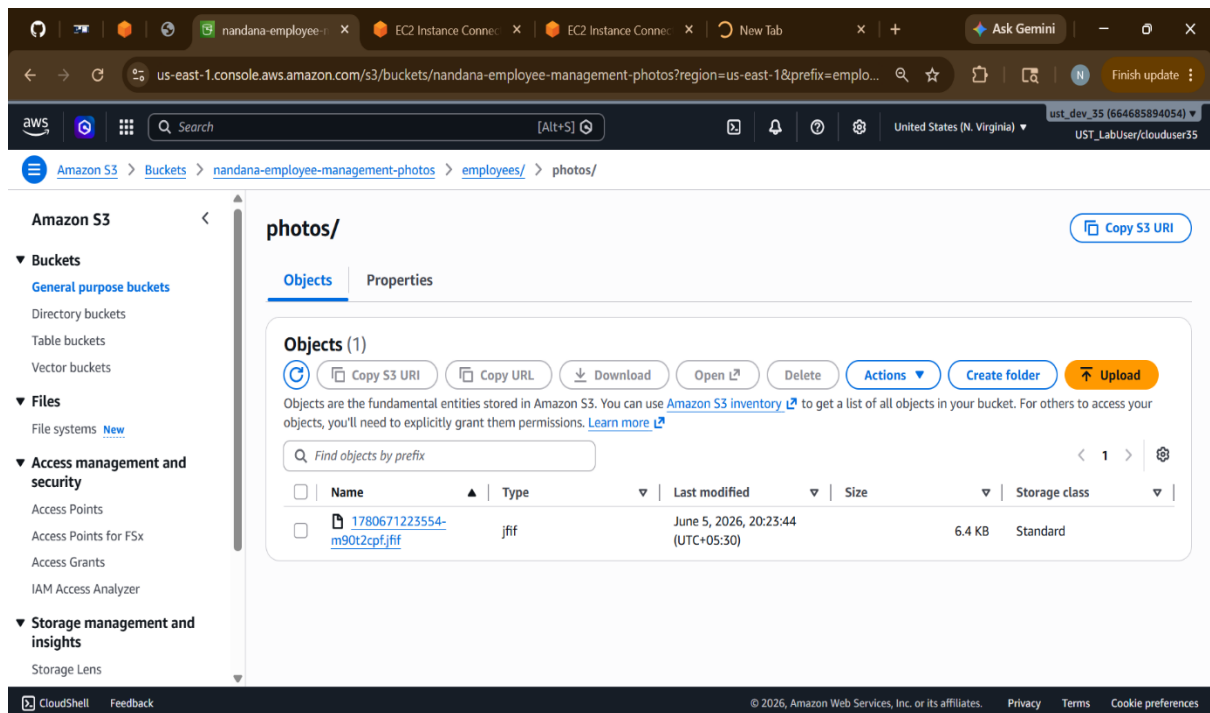
Implemented AWS Secrets Manager to securely store and retrieve MongoDB credentials, admin credentials, JWT secret, SMTP credentials. No credentials are hardcoded in the application.



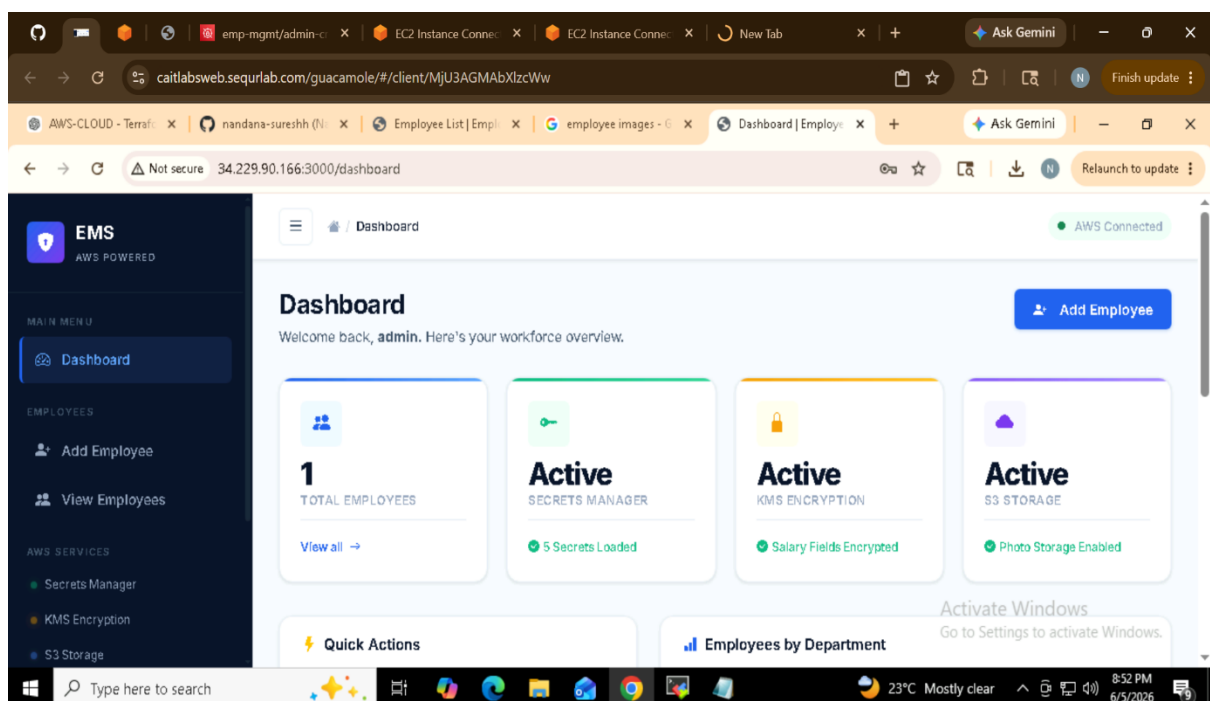
Implemented AWS KMS to encrypt employee salary data before storing it in MongoDB. Verified that only encrypted ciphertext is stored in the database and decrypted only when viewing employee details.



Implemented AWS S3 for employee profile photo storage. Employee images are uploaded to S3 and the object URL is stored in MongoDB.



Implemented JWT-based authentication using credentials retrieved from AWS Secrets Manager.



EMS
AWS POWERED

MAIN MENU

EMPLOYEES

AWS SERVICES

Dashboard

Add Employee

View Employees

Secrets Manager

KMS Encryption

S3 Storage

Employees / Add Employee

Add New Employee

Fill in the details below. The salary will be encrypted using AWS KMS and the photo will be stored in AWS S3.

Employee Information

Employee ID *

E.G. EMP001

Unique identifier (will be stored in uppercase)

Full Name *

e.g. John Smith

Email Address *

john.smith@company.com

Welcome email will be sent to this address

Phone Number *

+1 (555) 000-0000

Department *

Annual Salary (US\$)

Profile Photo

Click or drag to upload

Activate Windows
Go to Settings to activate Windows.

← Back to List

Type here to search23°C Mostly clear8:52 PM6/5/2026

EMS
AWS POWERED

MAIN MENU

EMPLOYEES

AWS SERVICES

Dashboard

Add Employee

View Employees

Secrets Manager

KMS Encryption

S3 Storage

Employees

Employee Directory

1 employees All salary data is encrypted at rest using AWS KMS

All Employees

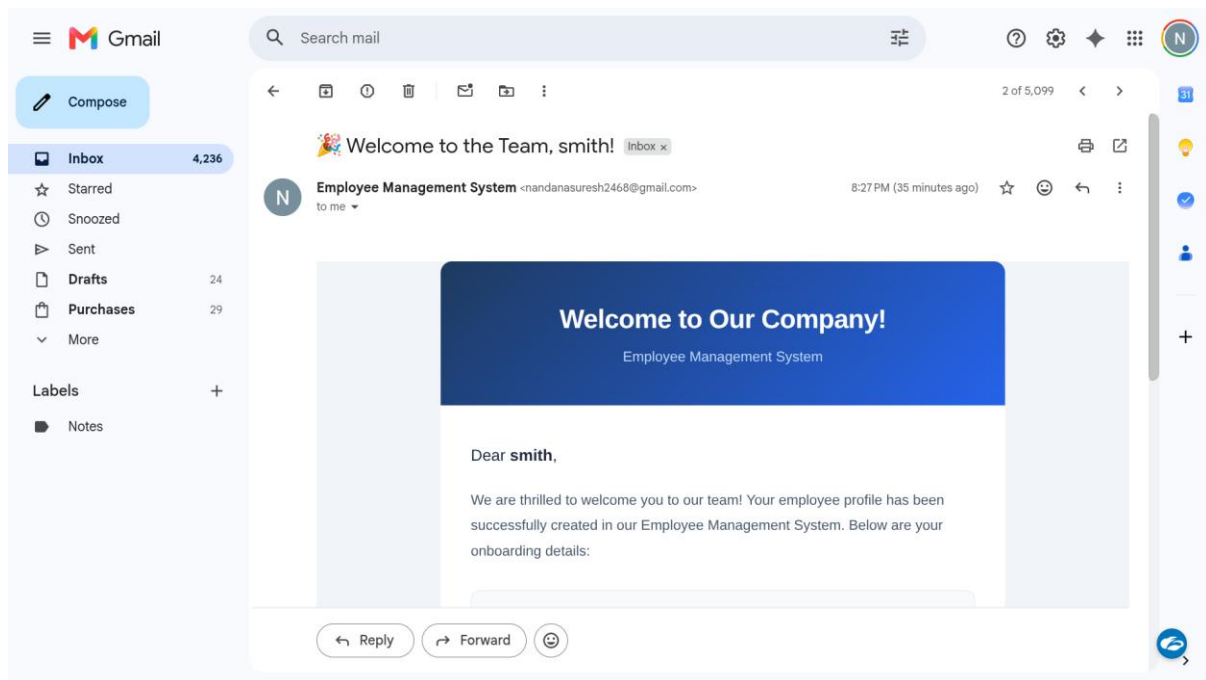
Search employees...

EMPLOYEE	EMPLOYEE ID	DEPARTMENT	EMAIL	PHONE	JOINED	ACTIONS
 john	ID: 1	Sales	john123@gmail.com	9789764567	Jun 5, 2026	View Delete

Activate Windows
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Configured SMTP email integration using credentials stored in Secrets Manager to send automated welcome emails when employees are created.



Deployed the application on EC2 and connected it to a MongoDB instance running on a separate EC2 server. Verified end-to-end functionality including login, employee creation, KMS encryption/decryption, S3 uploads, MongoDB storage, and email notifications.

Github repo url: <https://github.com/nandana-sureshh/emp-mgmt-system>